

Finishing look at time dependent perturbation theory

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We had our solution

$$c_1^1 = \frac{i\Gamma}{2\hbar} \left[\frac{e^{i(\omega+\gamma)t} - 1}{\omega + \gamma} - \frac{e^{i(\omega-\gamma)t} - 1}{\omega - \gamma} \right] \quad (1)$$

We get a driving force leading to an increase in amplitude and observe a beat frequency. When they are close together we have a slow oscillation and if they are far apart we get rapid oscillation. The width of our peak is given by approx $\frac{1}{\omega-\gamma}$

If we look at the first order coefficient for the ground state.

$$c_0^1(t) \quad (2)$$

Consider our ansatz.

$$|\psi(t)\rangle = (c_0^0 + g c_0^1(t)) |\psi_0\rangle e^{-i\omega_0 t} \quad (3)$$

What about the normalization? We will get a constraint. 'What do these solutions mean?' We want $\langle \psi(t) | \psi(t) \rangle = 1$. Our solution above may not necessarily be normalized g factor?

$$\langle \psi(t) | \psi(t) \rangle = (c_0^0 + g c_0^1(t))^* (c_0^0 + g c_0^1(t)) \quad (4)$$

$$= |c_0^0|^2 + g^2 \cdot \Re [c_0^{0*} c_0^1(t) + \mathcal{O}(g^2)] \quad (5)$$

Look out for symmetry e.g. parity check if functions are even for parity. In our case (drawn on board) we have even function so our g factor is zero. In our case our driving force is odd.

$$c_0^1 \propto \langle \psi_0 | \hat{V}(t) | \psi_0 \rangle = 0 \quad (6)$$

If we didn't have this normalization we would have to introduce a constraint. Due to the symmetry the normalization problem disappears.

Typically for physical systems we're interested in we have 'symmetric perturbations' e.g. electromag.

Switching off our perturbations we now go back to the infinite well.

For the static case we have

$$\psi_k = \sqrt{\frac{2}{a}} \sin\left(\frac{k\pi x}{2}\right) \quad (7)$$

$$E_k = \frac{k^2 \hbar^2 \pi^2}{2ma^2} = \hbar \omega_k = \hbar k^2 \lambda \quad (8)$$

If we now switch on our time dependence we get

$$\psi(x, t) = \sum_k c_k \psi_k(x) e^{-i\omega_k t} \quad (9)$$

$$\sum_{k=0}^{\infty} c_k = 1 \quad (10)$$

Example

$$\psi(x, t) = \frac{1}{\sqrt{a}} \left(\sin\left(\frac{\pi x}{a}\right) e^{-i\lambda t} + \sin\left(\frac{2\pi x}{a}\right) e^{-4i\lambda t} \right) \quad (11)$$

If we had put it into an Energy eigenstate we would get time independent Schrodinger equation which is $\propto \sin^2$
To find the prob at a position in the case where we have time dependence

$$|\psi(x, t)|^2 = \frac{1}{a} \left[\sin\left(\frac{\pi x}{a}\right) e^{-i\lambda t} + \sin\left(\frac{2\pi x}{a}\right) e^{-4i\lambda t} \right] \left[\sin\left(\frac{\pi x}{a}\right) e^{i\lambda t} + \sin\left(\frac{2\pi x}{a}\right) e^{4i\lambda t} \right] \quad (12)$$

$$= \frac{1}{a} \left[\sin^2\left(\frac{\pi x}{a}\right) + \sin^2\left(\frac{2\pi x}{a}\right) + 2 \sin\left(\frac{\pi x}{a}\right) \sin\left(\frac{2\pi x}{a}\right) \cos(3\lambda t) \right] \quad (13)$$

One could check unitarity

$$\int_0^a dx |\psi(x, t)|^2 = 1 \quad (14)$$

If we are in an energy eigenstate the time dependence won't effect the prob finding a particle at a place *in time*.